



**Pune District Education Association's Annasaheb Magar
Mahavidyalaya Hadapsar, Pune-411 028.**

DEPARTMENT OF ZOOLOGY

**FIELD PROJECT on
Territorial and Behavioral mapping of street dogs of
Manjari, Pune**

Submitted in partial fulfillment of the requirements
for the degree of Bachelor of Science(B.Sc.) in
Zoology

Submitted by

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Roll No: 3509

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Under the Guidance of

Dr. Sharad V Giramkar Submitted to

Department of Zoology

PDEA's Annasaheb Magar Mahavidyalaya Hadapsar, Pune-28

(Year 2026-27)

Affiliated to



Savitribai Phule Pune University

Pune 41100



**Pune District Education Association's Annasaheb Magar
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CERTIFICATE

This is to certify that the project report entitled “**Terrorial and Behavioral mapping of street dogs of Manjari, Pune**” has been successfully completed and submitted by **Ms.Kiran Bharat Makasare** (Exam. Seat No. 3509), a student of TY.B.Sc. Zoology (Major), for the academic year 2026–2027, under the supervision of Dr.Sharad V Giramkar, in partial fulfillment of the requirements for the degree of Bachelor of Science (B.Sc.) in Zoology (Major) under Savitribai Phule Pune University, Pune.

Project Guide
DR.Sharad Giramkar

Head of Department
(Signature and Stamp)



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Certificate by the Guide

This is to certify that the project work entitled "**Terrorial and Behavioral mapping of street dogs of Manjari, Pune**" submitted by **Ms.Kiran Bharat Makasare** (Exam Seat No 3509), of T.Y.B.Sc. Zoology (Major) for the academic year 2026–2027, has been completed under my guidance and supervision. The work embodied in this report is a genuine and original contribution carried out by the student as part of the undergraduate curriculum requirement.

Project Guide (Signature)

Name:.....

Designation:

Department of Zoology PDEA's Annasaheb Magar Mahavidyalaya, Hadapsar, Pune–28

Date:.....



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DECLARATION

I, **Ms. Kiran Bharat Makasare** (Exam Seat No:3509) student of T.Y.B.Sc. Zoology (Major) at PDEA's Annasaheb Magar Mahavidyalaya, Hadapsar, Pune-28, hereby declare that the project work entitled "**Terrorial and Behavioral mapping of street dogs of Manjari, Pune**" submitted to the Department of Zoology is the result of my original work carried out under the supervision of Dr. Sharad V Giramkar during the academic year 2026-2027. I further declare that this project has not been submitted to any other university or institution for the award of any degree or diploma.

Student's Signature

Name: Kiran Bharat Makasare

Roll No.: 3509

Date: _____

Place: Hadapsar, Pune



**Pune District Education Association's Annasaheb Magar
Mahavidyalaya Hadapsar, Pune-411 028.**

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Ms. Kiran Bharat Makasare

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CHAPTER 1: INTRODUCTION

1.1 Background

Free-ranging domestic dogs (*Canis lupus familiaris*) are commonly found in residential and semi-urban environments, where they interact with humans, other animals, roads, food resources and the surrounding environment. Their behaviour involves movement, social interaction, territorial activity and visible health conditions.

The present study, titled “**Spatial Mapping, Ethological Dynamics, and Open-Source Computational Tracking of Free-Ranging Canines (*Canis lupus familiaris*) in Residential Urban Ecosystems,**” combines field observation, behavioural assessment and spatial analysis. The study records demographic, behavioural, health and geographical information and analyses it using open-source computational tools, Python and GIS.

1.2 Free-Ranging Dogs in Urban Ecosystems

Free-ranging dogs interact with residential lanes, houses, roads, open spaces, food sources and human activities. Their distribution may change according to location and time.

The study records:

- Individual identification and sex
- Health-related conditions
- Observation time
- Human-related behavioural responses
- Defence roles
- Territorial interactions
- Geographical location

1.3 Importance of Spatial Mapping

Animal behaviour is closely associated with location. In free-ranging dogs, spatial distribution may be related to food availability, human activity, roads, resting areas and interactions with other dogs.

Therefore, longitude and latitude are included in the dataset to develop spatial maps and identify patterns of observed dog distribution.

1.4 Ethology and Behavioural Dynamics

Ethology is the study of animal behaviour under natural conditions. This project focuses on observable behaviour in residential environments.

The dataset includes **Human_Aggressiveness** and **Defence_Role** variables. Behaviour is considered as a response to a particular situation rather than a permanent characteristic of an individual dog.

1.5 Human–Dog Interactions

Free-ranging dogs frequently encounter human activities and environmental stimuli. The study records specific situations such as speeding bikes and children throwing stones and the corresponding canine responses.

This allows observations to be represented as:

Human/Environmental Stimulus → Observed Canine Response

1.6 Territorial Behaviour

The study uses lane-based locations such as **L1, L2, L3 and L4/5/6** to record territorial interactions. Variables such as **Is_Trespasser** and **Target_Territory** help document observed movement between areas.

These records represent observed territorial interactions and are not considered definitive ecological territory boundaries.

1.7 Time-Dependent Behaviour

The dataset includes different observation times ranging from early morning to night. These observations can be grouped into morning, afternoon, evening and night to identify patterns in recorded behaviour.

Since observations were made at selected time points, they represent patterns within the recorded dataset rather than the complete daily activity cycle.

1.8 Health and Welfare Assessment

Health assessment was based on visible field observations rather than veterinary diagnosis. The main recorded indicators were:

- **Ear_Cut**
- **Ticks**
- **Limping**

These observations can be used to develop a basic welfare profile and identify locations where health-related observations occur.

1.9 Demographic Characterization

Demographic information was organized using **Dog_Name, Dog_ID, Gender and Lane**. Individual identifiers allow repeated observations to be associated with particular dogs or groups during the study period.

1.10 Open-Source Computational Analysis

The project uses open-source computational methods to convert field observations into structured datasets, statistical summaries, behavioural graphs and spatial maps.

The general workflow is:

Field Observation → CSV Dataset → Data Cleaning → Python Analysis → Behavioural/Spatial Analysis → Graphs & GIS Maps → Interactive Output

The GitHub framework, “**Society Stray Dog Demographics & Behavioral Mapping**,” demonstrates the integration of demographic, behavioural, health and spatial information through Python and GIS-based visualization.

1.11 Problem Statement

Simple dog surveys generally focus on population counts. However, counting alone does not describe behavioural, spatial, territorial or health-related patterns.

Therefore, an integrated framework is required to determine:

- Who is present?
- Where are they located?
- When are they observed?
- How do they respond to humans and other dogs?
- What visible health indicators are present?
- Are territorial interactions observed?
- Can these observations be analysed computationally?

1.12 Aim of the Study

To investigate the spatial distribution, ethological dynamics, visible health indicators and territorial interactions of free-ranging domestic dogs (*Canis lupus familiaris*) in a residential urban ecosystem using field observations, geographical mapping and open-source computational analysis.

1.13 Objectives

1. To document the demographic characteristics of free-ranging dogs.
2. To identify individual dogs using names, IDs and locations.
3. To record behavioural responses to human and environmental stimuli.
4. To characterize observed defence roles.
5. To assess visible health indicators such as ear-cut status, ticks and limping.
6. To investigate spatial distribution using geographical coordinates.
7. To document observed territorial interactions.
8. To examine observations across different times of the day.
9. To analyse the dataset using Python-based computational methods.
10. To develop behavioural and GIS-based spatial visualizations.

1.14 Research Questions

1. What is the demographic composition of the recorded dogs?
2. How are the dogs spatially distributed?
3. What behavioural responses are associated with human or environmental stimuli?
4. What defence roles are observed?
5. What visible health conditions are recorded?
6. Are territorial crossings observed?
7. Do observations vary according to time of day?
8. Can computational analysis and GIS integrate behavioural, health and spatial observations?

1.15 Significance of the Study

The study integrates **ethology, spatial ecology, data management, Python-based analysis and GIS visualization.**

It provides:

- **Scientific significance:** links behaviour with location and time.
- **Computational significance:** demonstrates open-source analysis of field data.
- **Ecological significance:** maps spatial distribution and territorial interactions.
- **Welfare significance:** records visible health indicators.
- **Community significance:** documents human–dog interactions relevant to future animal-management studies.

CHAPTER 2: METHODOLOGY

The present study was carried out as a field-based observational study to understand the distribution, behaviour and visible health conditions of free-ranging domestic dogs (*Canis lupus familiaris*) in a residential urban environment. Observations were conducted across selected residential lanes identified as L1, L2, L3 and L4/5/6. The study focused on dog identification, behaviour, health, location and time of observation. Available geographical coordinates were recorded for spatial analysis.

During field observation, dogs were identified using visible characteristics, names and IDs. Information such as dog name, dog ID, gender, lane, observation time, geographical coordinates, human-associated behaviour, defence role and territorial interactions was recorded. Observations were made during different periods of the day, including morning, afternoon, evening and night. Repeated observations were retained to allow comparison of behaviour and location across different observation times.

Behavioural assessment focused on observable responses to human and environmental stimuli, including situations such as speeding bikes and children throwing stones. Defence roles such as front, secondary and defence were also recorded. Visible health observations included ear-cut status, tick presence and limping. Territorial behaviour was recorded using information on trespassing and target territory. These observations were treated as context-specific field observations rather than permanent behavioural characteristics or veterinary diagnoses.

Where available, latitude and longitude were recorded along with the corresponding dog observations. These geographical data were used to represent the distribution of dogs across residential lanes and to examine observed territorial relationships. The observation time was also recorded to allow comparison of recorded observations across different periods of the day.

After field collection, the observations were organized into a structured digital dataset. The data were checked for missing values, inconsistent dog names, time formats, coordinates and categorical responses. Missing information was retained as missing rather than being replaced with assumed values. The cleaned dataset was then processed using a Python-based open-source computational workflow to generate summaries, graphs and behavioural and health-related analyses.

The geographical information was further used for GIS-based spatial visualization and interactive mapping. The final workflow involved field observation, dog identification, demographic and behavioural assessment, health observation, time and location recording, data organization, cleaning, Python-based analysis and GIS visualization. The results were presented through tables, graphs, behavioural and health summaries, spatial maps and interactive visualizations.

CHAPTER 3: RESULTS, OBSERVATIONS AND ANALYSIS

The field observations recorded free-ranging dogs across residential lanes including L1, L2, L3 and L4/5/6. Identified dogs included Sweety, Cupcake, Goldy, Swara, Brown, d3 and d4, with both male and female dogs represented. Repeated observations of individuals such as Sweety and Cupcake allowed their characteristics and observations to be compared across different locations and times. The recorded data therefore provided a basic demographic and lane-wise representation of the observed dogs

Dog	Sex	Lane/Area
Sweety	Female	L1/L2
Cupcake	Male	L1/L2
Goldy	Male	L3
Swara	Female	L3
Brown	Male	L3
d3	Female	L4/5/6
d4	Male	L4/5/6

The observations were recorded at different times ranging from early morning to late night, including 6:30 AM, 8:00 AM, 9:00 AM, 2:45 PM, 3:30 PM, 5:30 PM, 9:30 PM, 11:00 PM and 12:30 AM. Behavioural responses were associated with specific human or environmental situations. For example, responses related to speeding bikes and children throwing stones were recorded. Defence roles such as front, secondary and defence were also observed among different dogs. These findings indicate that the recorded behavioural responses were linked to particular situations rather than representing permanent characteristics of the dogs.

Visible health-related observations were recorded using ear-cut status, tick presence and limping. Several dogs were recorded with ear cuts and ticks, while limping was observed in some individuals. Sweety was recorded with ear cut, ticks and minor limping, while Goldy was recorded with ear cut, ticks and limping. These observations provide basic information about visible health and welfare conditions within the recorded population. However, they represent field observations and should not be considered clinical diagnoses.

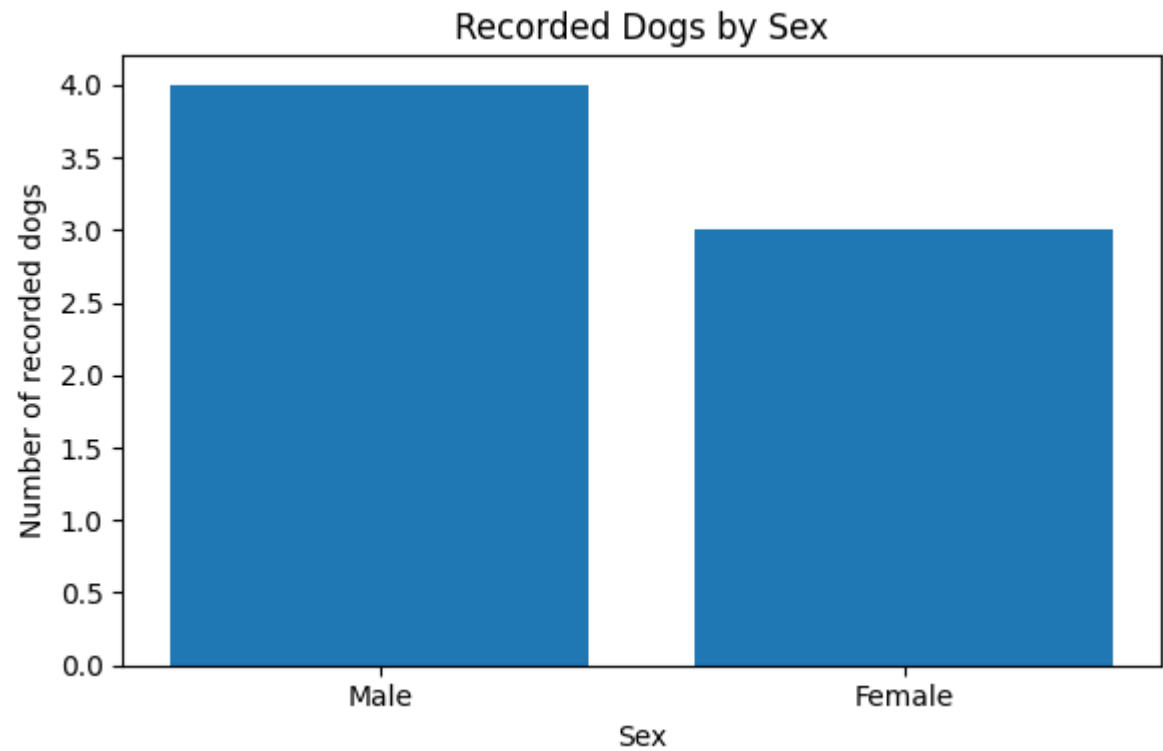
Spatial observations were obtained using recorded latitude and longitude values. The L1 observations were associated with approximately **73.984537 longitude and 18.519015 latitude**, while L3 observations were associated with approximately **73.984386 longitude and 18.518280 latitude**. Territorial interactions were recorded using the variables **Is_Trespasser** and **Target_Territory**. Some observations indicated movement or trespassing from L1 towards L3, providing evidence of observed interaction between different residential areas. These coordinates and territorial records were also used for GIS-based visualization.

The recorded demographic, behavioural, health, temporal and spatial information was organized into a structured dataset and processed through the project's computational workflow. Data cleaning and Python-based analysis were used to generate summaries, graphs and spatial visualizations, followed by an interactive GIS map. Overall, the observations demonstrated that the recorded dogs were distributed across different residential lanes, showed

context-specific behavioural responses, and exhibited visible health indicators that could be mapped according to location and time.

The results should be interpreted as **descriptive findings from the recorded observations**. Some entries were incomplete or contained variations in data formats; therefore, the findings represent patterns within the available dataset and should not be considered a complete estimate of the entire free-ranging dog population.

Dog	Ear cut	Ticks	Limping
Sweetie	Yes	Yes	Minor
Cupcake	Yes	Yes	No
Goldy	Yes	Yes	Yes
Swara	Yes	Yes	No
Brown	Yes	Yes	No



CHAPTER 4: DISCUSSION

The study showed that free-ranging dogs were distributed across different residential lanes, with repeated observations of identified dogs such as Sweetie, Cupcake, Goldy, Swara and Brown. Repeated identification helped distinguish individual dogs from separate sightings and provided information about their regular presence. The available GPS coordinates also helped connect observations with specific locations and supported spatial mapping of the study area.

The behavioural observations showed that canine responses were associated with specific environmental and human-related situations. Responses linked to speeding bikes and children throwing stones indicate that behaviour may vary according to the immediate situation. Defence roles such as front, secondary and defence were also recorded, suggesting differences in observed responses within groups. However, these roles should be considered observations during particular events rather than permanent social positions.

Human activity was an important part of the observed environment. Dogs were recorded at different times ranging from early morning to late night, providing an opportunity to compare observations across different periods. However, the data represent selected observation periods rather than continuous monitoring, so they should be interpreted as patterns within the recorded dataset.

Visible health indicators such as ticks, ear-cut status and limping were recorded in several dogs. These observations demonstrate that basic health and welfare information can be included in routine field surveys. Territorial interactions were also recorded, including observations involving trespassing from L1 towards L3. These records indicate observed interactions between residential areas but are not sufficient to define permanent territorial boundaries.

The combination of field observations, GPS information and computational analysis allowed demographic, behavioural, health, temporal and spatial data to be studied together. The Python and GIS-based workflow helped organize the data, identify inconsistencies and produce graphs and maps. Overall, the study demonstrates that a simple locally collected dataset can provide useful information about urban canine distribution, behaviour, health indicators and territorial interactions. Further studies with longer observation periods, more dogs and greater GPS coverage could provide more comprehensive results.

Appendix

